Question		Answer	Marks	AO	Guidance
16	(a)	A sample, since not all of the people in the village were selected	**B1** **[1]**	**2.4**	
16	(b)	H_0: $p = 0.0645$ H_1: $p \neq 0.0645$	**B1**	**1.1**	allow equivalent in words; do not allow percentages
		p is the probability that a person selected at random from the village is a vegetarian	**B1**	**2.5**	or p is the proportion of people in the village that are vegetarian **B1B1** if other symbol instead of p used if correctly defined
		Use or sight of B(78, 0.0645)	**M1***	**3.3**	
		P($X \geq 12$) = awrt 0.0042	**A1**	**1.1**	critical region is P($X \geq 11$) from 0.01134… and P($X \geq 10$) = awrt 0.02818 ignore calculation of lower tail, even where incorrect [and X = 0 from P($X \leq 1$) =0.03516 and P($X = 0$) = 0.0055]
		their 0.0042 correctly compared with 0.025 **or** their 0.0084 correctly compared with 0.05 do not accept H_0	**M1dep*** **A1FT**	**3.4** **1.1**	12 correctly compared with their 11 allow reject H_0 **or** accept H_1 or significant
		sufficient evidence to **suggest** at the 5% level that the probability that a person selected at random from the **village** is a vegetarian is **not** 0.0645	**A1** **[7]**	**2.2b**	dependent on award of all other marks apart from second **B1** do not allow eg conclude / prove / indicate or other assertive statement instead of suggest, accept use of not 6.45% rather than 0.0645
16	(c)	0.05	**B1** **[1]**	**1.2**	Allow P($X \geq 11$) + P($X = 0$) = 0.01134 + 0.0055 = 0.04099